Gerardo M. Casanola-Martin, Ph.D.

Research Associate | Coating and Polymeric Materials | North Dakota State University
Fargo, ND, USA gerardo.casanolamart@ndsu.edu

EDUCATION

Ph.D. in Chemistry (Cheminformatics) **2009**
Department of Chemistry, Central University of Las Villas (UCLV), Santa Clara, Cuba
Advisor: Prof. Yovani Marrero-Ponce

B.Sc. in Chemistry **2003**
Department of Chemistry, Central University of Las Villas (UCLV), Santa Clara, Cuba
Advisor: Prof. Lourdes Casas Cardoso

RESEARCH INTERESTS AND EXPERTISE

AI/ML for materials design, polymer informatics, graph neural networks, molecular simulation, quantum chemistry, and uncertainty-aware materials discovery; scientific computing; undergraduate and graduate research mentoring.

TEACHING EXPERIENCE

Teaching Assistant, Computational Characterization of Materials **2025; 2023**
Graduate course, Coating and Polymeric Materials program, North Dakota State University
- Assisted the instructor during hands-on laboratory sessions.

Assistant Professor **2015-2016**
Amazonian State University, Puyo, Ecuador
- General Chemistry, Organic Chemistry, and Analytical Chemistry (lectures and laboratories).

Assistant Professor **2014-2015**
PUCESE University, Esmeraldas, Ecuador
- General Chemistry, Organic Chemistry, and Analytical Chemistry (lectures and laboratories).

Instructor **2007-2010**
Central University of Las Villas (UCLV), Santa Clara, Cuba
- Bioorganic Chemistry I and Bioorganic Chemistry II (lectures and laboratories).

Instructor **2003-2007**
University of Ciego de Avila (UNICA), Ciego de Avila, Cuba
- General Chemistry and Analytical Chemistry (lectures and laboratories).

RESEARCH MENTORING

Graduate and Undergraduate Research Mentoring
North Dakota State University
- Mentor graduate researchers, including Ph.D. students, in computational chemistry, molecular modeling, data analysis, and research development.
- Provide scientific guidance to graduate mentors supervising undergraduate Summer Scholars, including research planning, interpretation of results, and research communication.

ACADEMIC AND RESEARCH APPOINTMENTS

Research Associate **May 2022-Present**
Coating and Polymeric Materials, North Dakota State University, Fargo, ND, USA
- Conduct data-science and cheminformatics studies related to bio-based and bio-renewable polymeric materials.
- Develop predictive models for polymer and materials properties and contribute to protocol and experimental-design planning.
- Implement Python workflows for automated data analysis and visualization in materials-science databases.

Senior Consultant / Freelance Data Scientist and Cheminformatics Specialist **May 2021-Present**
IKERDATA S.L.
- Develop perturbation-theory machine-learning (PTML) models and data-driven decision-support workflows.
- Apply artificial intelligence, statistics, complex networks, and cheminformatics to chemical and materials data.

Data Science and Cheminformatics Consultant **May 2019-May 2021**
Independent / Freelance
- Developed PTML models and applied AI, statistics, complex networks, and cheminformatics for decision-making models.

Visiting Research Assistant **Feb 2019-May 2019**
Coating and Polymeric Materials, North Dakota State University, Fargo, ND, USA
- Conducted cheminformatics and data-science studies of bio-based and bio-renewable polymeric materials.
- Developed Python workflows for data analysis, visualization, and polymer-property predictive modeling.

Data Science and Cheminformatics Consultant **Oct 2018-Jan 2019**
Independent / Freelance
- Performed computational screening of chemical systems and developed data-driven decision-making models.

Visiting Research Assistant **Apr 2018-Sep 2018**
Coating and Polymeric Materials, North Dakota State University, Fargo, ND, USA
- Conducted cheminformatics and data-science studies of bio-based and bio-renewable polymeric materials.
- Developed Python workflows for materials-science data analysis, visualization, and polymer-property prediction.

Senior Research Scientist **Mar 2017-Mar 2018**
University of Valencia, Valencia, Spain
- Developed chem-bioinformatics and multiscale machine-learning methodologies for computational discovery of enzyme inhibitors.
- Executed studies and prepared analytical and research reports.

Research Associate **Oct 2016-Jan 2017**
Department of Systems and Computer Engineering, Carleton University, Ottawa, Canada
- Performed feature engineering and advanced data analysis for chemoinformatics and bioinformatics methods addressing protein post-translational modifications and mutation-site prediction.

Assistant Professor **Sep 2015-Sep 2016**
Department of Life Sciences, Amazonian State University, Puyo, Ecuador

Assistant Professor **Oct 2014-Sep 2015**
School of Environmental Sciences, PUCESE University, Esmeraldas, Ecuador

Research Associate **Sep 2013-Mar 2014**
Biochemistry and Molecular Biology Department, University of Valencia, Spain
- Developed chem-bioinformatics and multiscale machine-learning methodologies for computational discovery of enzyme inhibitors.

Senior Research Scientist **Jan 2011-Sep 2013**
Center of Information and Technological Management, Ciego de Avila, Cuba
- Developed chem-bioinformatics, machine-learning, and multiscale computational-discovery methodologies.
- Served as a science advisor and consultant in drug design, information science, and technology.

Research Associate **Jul 2010-Dec 2010**
Biochemistry and Molecular Biology Department, University of Valencia, Spain
- Developed chem-bioinformatics and multiscale machine-learning methodologies for computational discovery of enzyme inhibitors.

Senior Research Scientist **Feb 2010-Jun 2010**
Computer Science Department, Bioplants Center, Ciego de Avila, Cuba
- Performed feature engineering and advanced statistical/machine-learning analysis for protein-protein interactions and contact-map prediction.

Research Assistant **Mar 2007-Jan 2010**
CAMD-BIR Unit and Department of Pharmaceutical Sciences, Central University of Las Villas, Cuba
- Performed data feature engineering, advanced statistical analysis, chem-bioinformatics, and multiscale machine-learning studies for computational discovery of enzyme inhibitors.

Visiting Researcher **Oct 2008-Dec 2008**
University Institute of Molecular Sciences, University of Valencia, Spain
- Scholarship for Young Researchers in Developing Countries; conducted chem-bioinformatics modeling for computational discovery of enzyme inhibitors.

RESEARCH FUNDING AND PROJECTS

- Ice Control Enhancement with Closed Loop Molecular Design (ICECycle). U.S. Defense Advanced Research Projects Agency (DARPA), Prime Award HR001124C0315; NDSU Subaward SUBK00018799, 2024–2026. Role: Researcher.
- Study of the Structure-Property Relationships of Coating Systems for Fouling Release. U.S. Office of Naval Research (ONR), Grant N00014-22-1-2129, 2022-2024. Role: Researcher.
- Data-Driven and Computationally Assisted Design of Near-Infrared Emissive Metal-Organic Complexes with Earth-Abundant Metals. U.S. Department of Energy, Grant DE-SC0022239, 2022-2024. Role: Researcher.
- Data-driven Computational and Machine Learning Assessment of Structure-Toxicity Relationship of Micro/NanoPlastics. EPSCoR RII: EPSCoR Research Fe, Grant 2229755, 2022-2025. Role: Researcher.
- NAFOSTED Research Project. Vietnam, Grant 108.05-2017.13, 2017-2018. Role: Co-Principal Investigator.
- Multi-scale Systems for Ecotoxicology Predictions of Different Biomarkers. Amazonian State University, UEA-2015, Sep 2015-Sep 2016. Role: Principal Investigator.
- Prediction of the Aquatic Toxicity of Organic Compounds Using Ensemble Systems Based on Artificial and Statistical Techniques. PUCESE University, CID-PUCESE-2015, Feb 2015-Aug 2015. Role: Principal Investigator.
- Strengthening Postgraduate Education and Research in Pharmaceutical Sciences. VLIR-UCLV IUC Program, Flemish Interuniversity Council, Belgium, 2008-2013. Role: Researcher.

SELECTED AWARDS, FELLOWSHIPS, AND PRESENTATIONS

- Invited Speaker, Multidisciplinary Research Symposium, Institute of Technological Development, UTEM, Chile.
- ELAP Scholarship, Global Affairs Canada, 2016.

- Workshop talk: “Proteasome: A promissory pharmacological target for the discovery of new chemical with anti-cancer activity: The virtual screening strategy for the search of novel inhibitors.” University of Valencia, Spain, March 14, 2014.
- Conference talk: “Novel Chemoinformatics Models for the Discovery of Potent Tyrosinase Enzyme Inhibitors.” University of the Basque Country, Spain, November 11, 2013.
- Maximum Award, National Academy of Sciences of Cuba, 2007 and 2013.
- Conference talk: “Integration of in-silico and in-vitro approaches for the discovery of novel tyrosinase inhibitor compounds.” Bioplants Center, University of Ciego de Avila, Cuba, March 11, 2011.
- Postdoctoral Fellowship, University of Valencia, 2010 and 2013-2014.
- Award for Young Researchers, University of Valencia, 2008.

PROFESSIONAL SERVICE AND MEMBERSHIPS

- Editorial Board Member, Current Issues in Molecular Biology (MDPI), 2026-present.
- Guest Editor, Molecules, Special Issue: "Density Functional Theory (DFT): From Conceptual Developments to Practical Applications," 2026.
- Organizing Committee Member, MOL2NET Conferences, 2015-present.
- Member, American Chemical Society, 2023-present.
- Member, American Society for Computational and Cellular Toxicology, 2023-present.
- Peer reviewer for more than 50 manuscripts, including submissions to Environmental Science and Ecotechnology, Science of the Total Environment, ACS Omega, Chemometrics and Intelligent Laboratory Systems, Medicinal Chemistry Communications, Molecular Diversity, Molecular Informatics, Molecular Simulation, ChemMedChem, and Journal of Materials Chemistry A, among others.

LANGUAGES

Spanish: Native
English: Full professional proficiency
Italian: Basic

BOOK CHAPTERS

1. Casañola-Martin, G.M., Pham-The, H. Machine learning applications in nanomedicine and nanotoxicology: An overview. In: Research Anthology on Machine Learning Techniques, Methods, and Applications, 2022, pp. 38–45
2. Casañola-Martín, G.M., Le-Thi-Thu, H.; Marrero-Ponce, Y.; Torrens, F.; Rescigno, F.; Abad, C.; Khan, M.T.H. QSAR-based Classification Models and TOMOCOMD-CARD Approach for the Discovery of New Tyrosinase Inhibitors. In: Recent Trends on QSAR in the Pharmaceutical Perceptions, Mahmud Tareq Hassan Khan, (Editor). Bentham Sciences Publishers. 2012, 296-337.

JOURNAL ARTICLES

79 journal articles, listed in reverse chronological order.

79. Nguyen Thi Thu Ha, Hai Pham-The, Dao Long Vu, Nguyen Van Tuyen, Phan Thu Huyen, Nguyen Thi Thuy Linh, Bui Hai Ninh, Le Thi Tu Anh, Ba Thi Cham, Nguyen Thanh Tra, Nguyen Thi Van Anh, Bakhtiyor Rasulev, Gerardo M. Casanola-Martin. Phytochemical Constituents from Ardisia sylvestris Pitard Leaves: Enzyme Inhibition, Antioxidant, and Antimicrobial Activities with Molecular Simulation Insights. Current Issues in Molecular Biology, 2026, 48(10), 986. DOI: 10.3390/cimb48100986.
78. Juan A. Castillo-Garit, Josue Pozo-Martinez, Esteban Rocha-Valderrama, Cristian Rojas-Peña, Karen Acosta-Quiroga, Vicente J. Arán, Claudio Olea-Azar, Gerardo M. Casañola-Martin, Bakhtiyor Rasulev, Francisco Torrens, Facundo Pérez-Giménez, Mauricio Moncada-Basualto. Dry Selection and Wet Evaluation of New 1,2-Disubstituted Nitroindazolin-3-One Derivatives as Promising Agents Against Trypanosoma cruzi. Pharmaceuticals, 2026, 19(9), 1473. DOI: 10.3390/ph19091473.
77. Gerardo M. Casanola-Martin, Estefania Ascencio, Ayda Dadras, Yehor Polunin, Seyedali Mirmohammadsadeghi, Humbert González-Díaz, Sergiy Minko, Andriy Voronov, Bakhtiyor Rasulev. Water Permeability Coefficient Prediction for Polymers applying Explainable Machine Learning. Macromolecules, 2026, 59(13), 7323-7336.
76. Kamrun N. Keya, Amirreza Daghighi, Gerardo M. Casanola-Martin, Wenjie Xia, Bakhtiyor Rasulev. Comparative Evaluation of Machine Learning-based QSPR Techniques for Predicting Polymer Glass Transition Temperature. Discover Materials, 2026, 6(1), 154.
75. Durbek Usmanov, Priyanka Yadav, Gerardo M. Casanola-Martin, Akshat Mallya, Seyedehelham Shirvanihosseini, Allison Hubel, Bakhtiyor Rasulev. Machine Learning Framework to Predict Glass Transition Temperature in Natural Deep Eutectic Solvents: A Step toward Green Functional Materials. Green Chemistry, 2026, 28(23), 9650-9666.
74. Thi-Van-Anh Nguyen, Duong Quang Quy, Nguyen Thanh Tung, Nguyen Thu Huyen , Pham Le Minh, Nguyen T. Minh Huyen, Tue-Tam Ho, Nguyen Thi Thu Ha, Gerardo M. Casanola-Martin, Bakhtiyor Rasulev, Hai Pham-The. Beyond α-Glucosidase and α-Amylase Inhibition: Integrated In Vitro and Multi-Scale In Silico Insights into the Antidiabetic and Antioxidant Mechanisms of Oxalis corniculata L. Aerial Parts. Molecules, 2026, 31, 630.
73. Seyedehelham Shirvanihosseini, Joan Manuel Rodríguez-Díaz, Gerardo M. Casanola-Martin, Bakhtiyor Rasulev. Machine learning based assessment of the toxicity of polymeric materials to green algae as an approach to design non-toxic polymeric coatings to deter marine biofouling. Results in Chemistry, 2026, 20, 103005.
72. José L. Rodríguez, Estefania Ascencio, José M. Ageitos, Lucia Feijoo, Shan He, Amirreza Daghighi, Gerardo M. Casanola-Martin, Cristian R. Munteanu, Santiago Rodríguez Yanez,* Alejandro Pazos, Harbil Bediaga, Brenda Durand Ordiales, Raquel Esteban, Ana-María Heres, Jorge Curiel Yuste, Lur Epelde, Bakhtiyor Rasulev, Tomas González, Sonia Arrasate, Humberto González-Diaz. Polymerase Chain Reaction. Perturbation Theory and Machine Learning Artificial Intelligence-Experimental Microbiome Analysis: Applications to Ancient DNA and Tree Soil Metagenomics Cases of Study. Advanced Intelligent Systems, 2026, e202500587.
71. Karel Dieguez-Santana, Gerardo M Casanola-Martin, Roldan Torres-Gutiérrez, Bakhtiyor Rasulev, Humberto Gonzalez-Diaz. AQUA Tox: A web tool for predicting aquatic toxicity in rotifer species using intrinsic explainable models. Journal of Hazardous Materials, 2025, 492, 138050.

70. Karel Diéguez-Santana, Gerardo M Casanola-Martin, Roldan Torres-Gutiérrez, Bakhtiyor Rasulev, Humberto González-Díaz. First report on Quantitative Structure-Toxicity Relationship modeling approaches for the prediction of acute toxicity of various organic chemicals against rotifer species. Science of The Total Environment, 2025, 997, 179350.
69. Rahil Ashtari Mahini, Gerardo Casanola-Martin, Stephen Szwiec, Simone A Ludwig, Bakhtiyor Rasulev. CombinatorixPy: Advancing mixture descriptors for computational chemistry. SoftwareX, 2025, 29, 102060.
68. Durbek Usmanov, Ugiloy Yusupova, Vladimir Syrov, Gerardo M. Casanola-Martin, Bakhtiyor Rasulev. A Quantitative Structure–Activity Relationship Study of the Anabolic Activity of Ecdysteroids. Computation, 2025, 13, 195.
67. Melanie Ochoa-Ocampo, Nina Espinosa de los Monteros-Silva, Jefferson V Pastuña-Fasso, Juan Diego Sacoto, María Cristina Peñuela-Mora, Gerardo Casanola-Martin, José R Almeida, Karel Diéguez-Santana, Noroska GS Mogollón. Volatile Compositional Profile, Antioxidant Properties, and Molecular Docking of Ethanolic Extracts from Philodendron heleniae. Molecules, 2025, 30(6), 1366.
66. Shan He, Ander Baron, Cristian R Munteanu, Begoña de Bilbao, Gerardo M Casanola-Martin, Mariana Chelu, Adina Magdalena Musuc, Harbil Bediaga, Estefania Ascencio, Idoia Castellanos-Rubio, Sonia Arrasate, Alejandro Pazos, Maite Insausti, Bakhtiyor Rasulev, Humberto Gonzalez-Diaz. Drug Release Nanoparticle System Design: Data Set Compilation and Machine Learning Modeling. ACS Applied Materials & Interfaces, 2025, 17(3) 5290-5306.
65. Casanola-Martin G.M., Wang J., Zhou J., Rasulev B., Leszczynski J. Chemical Feature-Based Machine Learning Model for Predicting Photophysical Properties of BODIPY Compounds: Density Functional Theory and Quantitative Structure–Property Relationship Modeling, Journal of Molecular Modeling, 2025 31(1), 18.
64. Meringer M., Casanola-Martin G.M., Rasulev B., Cleaves II H.J. Similarity Analysis of Computer-Generated and Commercial Libraries for Targeted Biocompatible Coded Amino Acid Replacement, International Journal of Molecular Sciences, 2024, 25(22), 12343.
63. Gerardo M. Casanola-Martin, Hai Pham-The, Humbert González-Díaz, Dean C. Webster, Bakhtiyor Rasulev. Machine Learning Analysis of a Large Set of Homopolymers to Predict Glass Transition Temperatures. Communications Chemistry. 2024, 7, 226.
62. Rahil Ashtari Mahini, Gerardo M. Casanola-Martin, Simone A. Ludwig, Bakhtiyor Rasulev. MixtureMetrics: A comprehensive package to develop additive numerical features to describe complex materials for machine learning modeling. SoftwareX. 2024. 28, 101911.
61. Tetiana Shevtsova, Kweeni Iduoku, Kristen Patnode Setien, Inioluwa Olabode, Gerardo M. Casanola-Martin, Bakhtiyor Rasulev, Andriy Voronov. Enhancing Plant Proteins-Based Bioplastics with Natural Additives: A Comprehensive Study by Experimental and Computational Approaches. ACS Sustainable Chemistry & Engineering. 2024, 12(43), 15948-15960.
60. Prediction of Dielectric Constant in Series of Polymers by Quantitative Structure-Property Relationship (QSPR). Estefania Ascencio-Medina, Shan He, Amirreza Daghighi, Kweeni Iduoku, Gerardo M. Casanola-Martin, Sonia Arrasate, Humberto González-Díaz and Bakhtiyor Rasulev. Polymers 2024, 16(19), 2731
59. NANO.PTML model for read-across prediction of nanosystems in neurosciences. computational model and experimental case of study. Shan He, Karam Nader, Julen Segura Abarrategi, Harbil Bediaga, Deyani Nocedo-Mena, Estefania Ascencio, Gerardo M. Casanola-Martin, Idoia Castellanos-Rubio, Maite Insausti, Bakhtiyor Rasulev, Sonia Arrasate, Humberto González-Díaz. Journal of Nanobiotechnology (2024) 22:435.
58. Design and Evaluation of Nanoscale Materials with Programmed Responsivity towards Epigenetic Enzymes. Priyanka Ray, Abbas Sedigh, Matthew Confeld, Lina Alhalhooly, Kweeni Iduoku, Gerardo M. Casanola-Martin, Hai Pham-The, Bakhtiyor Rasulev, Yongki Choi, Zhongyu Yang, Sanku Mallik, Mohiuddin Quadir. Journal of Materials Chemistry B, 2024, 12(39), 9905-9920.
57. Iduoku, K., Ngongang, M., Kulathunga, J., Daghighi, A., Casanola-Martin, G. M., Simsek, S., Rasulev, B. Phenolic Acid–β-Cyclodextrin Complexation Study to Mask Bitterness in Wheat Bran: A Machine Learning-Based QSAR Study. Foods, 2024, 13(13), 2147
56. Amirreza Daghighi, Gerardo M. Casanola-Martin, Kweeni Iduoku, Hrvoje Kusic, Humberto Gonzalez-Diaz, and Bakhtiyor Rasulev. Multi-Endpoint Acute Toxicity Assessment of Organic Compounds Using Large-Scale Machine Learning Modeling. Environmental Science and Technology, 2024, 58(23), pp. 10116–10127.
55. Anas Karuth, Stephen Szwiec, Gerardo M. Casanola-Martin, Achiya Khanam, Maryam Safaripour, David Boucher, Wenjie Xia, Dean C. Webster, Bakhtiyor Rasulev. Integrated machine learning, computational, and experimental investigation of compatibility in oil-modified silicone elastomer coatings. Progress in Organic Coatings, 193 (2024) 108526.
54. Meade Erickson, Gerardo M. Casañola-Martin, Yulun Han, Bakhtiyor Rasulev, and Dmitri Kilin. Relationships between the Photodegradation Reaction Rate and Structural Properties of Polymer Systems. Journal of Physical Chemistry B, 2024, 128(9), pp. 2190–2200
53. Yevhenii Zhuravskyi, Kweeni Iduoku, Meade E. Erickson, Anas Karuth, Durbek Usmanov, Gerardo M. Casanola-Martin, Maqsud N. Sayfiyev, Dilshod A. Ziyaev, Zulayho Smanova, Alicja Mikolajczyk, Bakhtiyor Rasulev. Quantitative Structure-Permittivity Relationship Study of a Series of Polymers. ACS Materials Au. 2024, 4(2), 195-203.
52. Anas Karuth, Gerardo M. Casanola-Martin, Levi Lystrom, Wenfang Sun, Dmitri Kilin, Svetlana Kilina, and Bakhtiyor Rasulev. Combined Machine Learning, Computational, and Experimental Analysis of the Iridium(III) Complexes with Red to Near-Infrared Emission. J. Phys. Chem. Lett. 2024, 15, 471−480.
51. Sarduy-Pereira, L.B., Elisabeth De Decker, M.K., Pérez-Martínez, A., Casañola-Martin, G.M., Diéguez-Santana, K. The socio-economic and environmental impacts on gravel mining in rivers: a case study in the Ecuadorian Amazon Region. International Journal of Sustainable Society, 2024, 16(1), pp. 1–20.
50. Phan Thi Diem-Tran, Tue-Tam Ho, Nguyen-Van Tuan, Le-Quang Bao, Ha Tran Phuong, Trinh Thi Giao Chau, Hoang Thi Binh Minh, Cong-Truong Nguyen, Zulayho Smanova, Gerardo M. Casanola-Martin, Bakhtiyor Rasulev, Hai Pham-The, Le Canh Viet Cuong Stability Constant and Potentiometric Sensitivity of Heavy Metal–Organic Fluorescent Compound Complexes: QSPR Models for Prediction and Design of Novel Coumarin-like Ligands. Toxics, 2023, 11(7), 595
49. Le-Quang Bao, Phuong Linh Nguyen, Nguyen-Hai Nam, Nguyen Thi Thuan, Phan Thi Phuong Dung, Bakhtiyor Rasulev, Gerardo M. Casanola-Martin, Hai Pham-The. Development of Activity Rules and Chemical Fragment Design for In Silico Discovery of AChE and BACE1 Dual Inhibitors against Alzheimer's Disease. Molecules, 2023, 28(8), 3588
48. Martínez-López, Y., Castillo-Garit, J.A., Casanola-Martin, G.M., Rasulev, B., Martínez-Santiago, O., Barigye, S.J . Exploring proteasome inhibition using atomic weighted vector indices and machine learning approaches. Molecular Diversity, 2023.

47. Amirreza Daghighi, Troy Timmerman, Gerardo M. Casanola-Martin, Dejan Milenkovic, Bono Lucic, Bakhtiyor Rasulev. In Silico Prediction of the Toxicity of Nitroaromatic Compounds: Application of Ensemble Learning QSAR Approach. Toxics. Toxics, 2022, 10(12), 746
46. Karel Diéguez-Santana, Amilkar Puris, Oscar M. Rivera-Borroto, Gerardo M. Casanola-Martin, Bakhtiyor Rasulev, and Humberto González-Díaz. A Classification Fuzzy System Approach for QSAR Modeling of α-Amylase and α-Glucosidase Inhibitors. Current Computer Aided Drug Design. 2022, 18(7), pp. 469–479
45. Karabulut, S., Mansour, B., Casanola-Martin, G.M., Rasulev, B., Gauld, J.W. The Hydrolysis Rate of Paraoxonase-1 Q and R Isoenzymes: An In Silico Study Based on In Vitro Data. Molecules, 2022, 27(20), 6780
44. Karel Diéguez-Santana, Gerardo M. Casañola-Martin, Roldan Torres, Bakhtiyor Rasulev, James R. Green and Humbert González-Díaz. Machine Learning Study of Metabolic Networks vs ChEMBL Data of Antibacterial Compounds. Mol. Pharmaceutics. 2022, 19, 7, 2151–2163.
43. Karel Diéguez-Santana, Gerardo M. Casañola-Martin, James R. Green, Bakhtiyor Rasulev, Humberto González-Díaz. Predicting Metabolic Reaction Networks with Perturbation-Theory Machine Learning (PTML) Models. March 2021. Current Topics in Medicinal Chemistry 21(9). DOI: 10.2174/1568026621666210331161144.
42. Bakhtiyor Rasulev, Gerardo M. Casañola-Martin. QSAR/QSPR in Polymers: Recent Developments in Property Modeling. International Journal of Quantitative Structure-Property Relationships . 2020, 5(1):80-88. DOI: 10.4018/IJQSPR.2020010105.
41. Karel Diéguez-Santana, Oscar M. Rivera-Borroto, Amilkar Puris, Hai Pham-The, Huong Le-Thi-Thu, Bakhtiyor Rasulev, Gerardo M. Casañola-Martin. Beyond Model Interpretability using LDA and Decision Trees for α-Amylase and α-Glucosidase Inhibitor Classification Studies. Chemical Biology and Drug Design. 2019. DOI: 10.1111/cbdd.13518.
40. Pham-The H, Cabrera-Pérez M, Nam NH, Castillo-Garit JA, Rasulev B, Le-Thi-Thu H, Casañola-Martin GM. In Silico Assessment of ADME Properties: Advances in Caco-2 Cell Monolayer Permeability Modeling. Curr Top Med Chem. 2018, 18(26), 2209-2229.
39. Hai Pham-The, Nguyen-Hai Nam, Doan-Viet Nga , Dang Thanh Hai, Karel Dieguez-Santana, Yovani Marrero-Ponce, Juan A. Castillo-Garit, Gerardo M. Casañola-Martin, Huong Le-Thi-Thu. Learning from Multiple Classifier Systems: Perspectives for Improving Decision Making of QSAR Models in Medicinal Chemistry. Current Topics in Medicinal Chemistry. 2018, 17(30),3269-3288
38. Gerardo M. Casañola-Martin, Hai Pham-The, Juan A. Castillo-Garit and Huong Le-Thi-Thu. Atom based Linear Index Descriptors in QSAR-Machine Learning Classifiers for the Prediction of Ubiquitin-Proteasome Pathway Activity. Medicinal Chemistry Research. 2018. 27(3), 695-704.
37. Michael González-Durruthy, Jose M. Monserrat, Bakhtiyor Rasulev, Gerardo M. Casañola-Martín, José María Barreiro Sorrivas, Sergio Paraíso-Medina, Víctor Maojo, Humberto González-Díaz, Alejandro Pazos and Cristian R. Munteanu. Carbon Nanotubes' Effect on Mitochondrial Oxygen Flux Dynamics: Polarography Experimental Study and Machine Learning Models using Star Graph Trace Invariants of Raman Spectra. Nanomaterials. 2017 7(11), 386.
36. Juan. A. Castillo-Garit, Gerardo. M. Casañola-Martin, Stephen. J. Barigye, Hai. Pham-The, Francisco Torrens & Amparo. Torreblanca. Machine learning-based models to predict modes of toxic action of phenols to Tetrahymena pyriformis. SAR and QSAR in Environmental Research. 2017, 28(9), 735-747
35. Hai Pham-The, Gerardo M. Casañola-Martin, Karel Dieguez-Santana, Nam Nguyen Hai, Nguyen Thi Ngoc, Loi Vu Duc, Huong Le-Thi-Thu. Quantitative structure–activity relationship analysis and virtual screening studies for identifying HDAC2 inhibitors from known HDAC bioactive chemical libraries. SAR and QSAR in Environmental Research, 2017, 28(3), 199-220.
34. Karel Dieguez-Santana, Hai Pham-The, Oscar M. Rivera-Borroto, Amilkar Puris, Huong Le-Thi-Thu, Gerardo M. Casañola-Martin. A Two QSAR Way for Antidiabetic Agents Targeting using α-Amylase and α-Glucosidase Inhibitors: Model Parameters Settings in Artificial Intelligence Techniques. Letters in Drug Design & Discovery. 2017, 14(8), 862 – 868
33. Juan A. Castillo-Garit, Gerardo M. Casañola-Martin, Huong Le-Thi-Thu, Hai Pham-The and Stephen Jones-Barigye A Simple Method to Predict Blood-Brain Barrier Permeability of Drug-Like Compounds Using Classification Trees. Medicinal Chemistry. 2017. 13, 1-6.
32. Pedro Julio Villegas, Gerardo M. Casañola-Martin, and Karel Diéguez-Santana. Prediction of the adsorption capacity of activated carbons using physicochemical parameters, mechanical properties and elemental composition. International Journal of ChemTech Research. 2016, 9, 740-748.
31. Karel Dieguez-Santana, Hai Pham-The, Pedro J. Villegas-Aguilar, Huong Le-Thi-Thu, Juan A. Castillo-Garit, Gerardo M. Casañola-Martin. Prediction of acute toxicity of phenol derivatives using multiple linear regression approach for Tetrahymena pyriformis contaminant identification in a median-size database. Chemosphere. 2016. 165, 434-441.
30. Juan A. Castillo-Garit, Concepción Abad, Gerardo M. Casañola-Martin, Stephen Jones-Barigye, Francisco Torrens and Amparo Torreblanca. Prediction of Aquatic Toxicity of Benzene Derivatives to Tetrahymena pyriformis according to OECD principles. Current Pharmaceutical Design. 2016. 22(33), 5085-5094.
29. Gerardo M. Casañola-Martin, Huong Le-Thi-Thu, Facundo Pérez-Giménez, Yovani Marrero-Ponce, Matilde Merino-Sanjuán, Concepción Abad, and Humberto González-Díaz. Prediction of Drug–Target Interactions for ubiquitin-proteasome system inhibitors using Bob-Jenkins Operators of TOMOCOMD quadratic indices. Current Protein and Peptide Sciences. 2016. 17(3), 220-227.
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